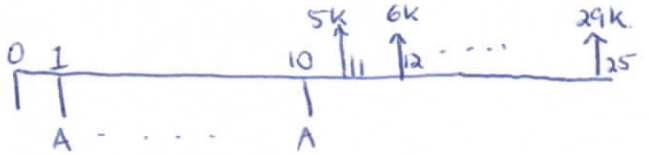
ANSWER KEY (MIDTERM 1)

1) (d) $ROE = \dfrac{\text{Net Income}}{\text{Shareholder Equity}} = \dfrac{\overbrace{(215-10)\times(1-.32)}^{\text{After Tax net Income}}}{700-280} = .33$

2) (b)



$G = 1000$
$r = 6\%$

$$V_{10} = A\times\underbrace{(F|A,6\%,10)}_{13.1802} = 5000\underbrace{(P|A,6\%,15)}_{9.7122} + 1000\underbrace{(P|G,6\%,15)}_{57.5546}$$

$$A = 8050.77$$

3) (e)

$$P + 500\overset{.5132}{(P|F,10\%,7)} \geq 1000\overset{6.1446}{(P|A,10\%,10)} - 1000\overset{.5132}{(P|F,10\%,7)}$$

$$P \geq 5374.8$$

4) (d) is incorrect

5) (a)

$$100\times\overset{3.3100}{(F|A,10\%,3)}\overset{1.10}{(F|P,10\%,1)} + 50\overset{2.4869}{(P|A,10\%,3)} = F_3$$

$$F_3 = 488.45$$

6) (c)

$$C + C\overset{4.8684}{(P|A,10\%,7)} = \left[1000\overset{3.1699}{(P|A,10\%,4)} + 200\overset{4.3781}{(P|G,10\%,4)}\right]\overset{(.7513)}{(P|F,10\%,3)}$$

$$C(1+4.8684) = 3039.4$$

$$C = 517.93$$

7) (e)

$$V = 100\overset{6.3528}{(F|A,12\%,5)} + 108\underbrace{\overset{4.1566}{(P|108,8\%,12\%,5)}}_{\text{formula: } \left[\frac{1-(1.08)^5(1.12)^{-5}}{.12-.08}\right]}$$

$$= 1084.19$$

8) (d)

$$I_2 = 1000(1.09)^2 - 1000(1.09)$$
$$B_1(1.09) - B_1$$
$$= 98.1$$

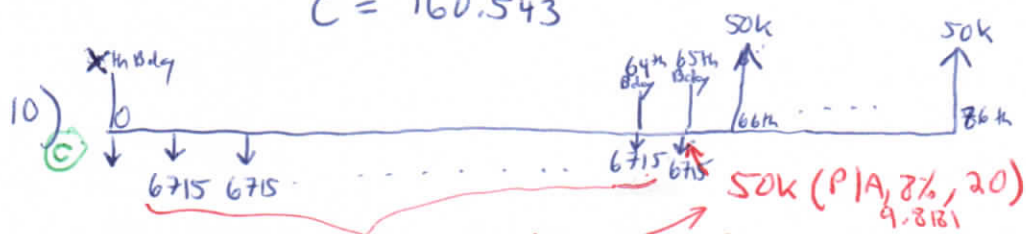
$\left.\right\}$ $yr.1 = 1000\times1.09 = 1090$
$yr.2 = 1000\times(1.09)^2 - 1090 = 98.1$

9) (b)

$$V_1 = C\overset{1.1}{(F|P,10\%,1)} + C\overset{3.1699}{(P|A,10\%,4)} = 100\overset{1.1}{(F|P,10\%,1)} + 100 + 150\overset{3.1699}{(P|A,10\%,4)}$$

$$C = 160.543$$

10) (c)



$50k\,(P|A,8\%,20)$
9.8181

$6715(F|A,8\%,Y)$ $\longleftrightarrow$ $Y(F|A,8\%,Y) = 73.1057$

$Y \approx 25$ years $\Rightarrow X = 65 - 25 = 40$th Bdy.

11) (c) $100 = 115(P|F, i, 2)$

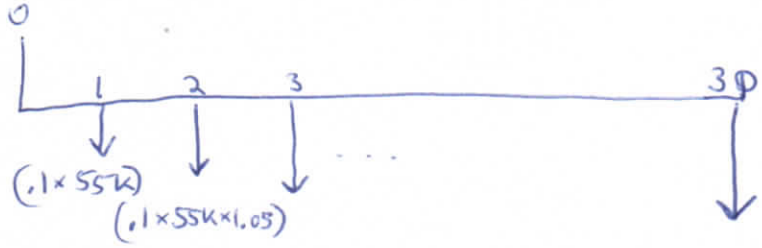
$$100 = 115\left(\frac{1}{1+i}\right)^2$$

$$(1+i)^2 = 1.15$$

$$1+i = 1.07238$$

$$i = .07238.$$

12) (e)



$g = 5\%$

$i = 9\%$

$$(55K \times .1)\underbrace{(P|55K\times.1, 5\%, 9\%, 30)}_{16.8563}(F|P, 9\%, 30) = \$1{,}230{,}040.71$$

$$\left[\frac{1-(1.05)^{30}(1.09)^{-30}}{.09-.05}\right]$$

13.2677

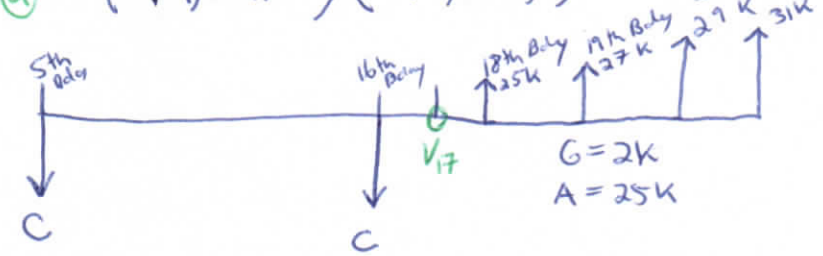
13) (a) $C \times (F|A, 9.5\%, 12)(F|P, 9.5\%, 1) = V_{17}$

20.7522 1.095

$V_{17} = 25K\,(P|A, 9.5\%, 4) + 2K\,(P|G, 9.5\%, 4)$

3.2045

4.4440



$\Rightarrow C = 3916.64$

14) (a) $i_m = \frac{.125}{12} = .01042$

$$2000(1.01042)^2 = \$2041.88$$

15) (e) $1^{\$}\left(1 + \frac{r}{12}\right)^{60} = 5^{\$}$

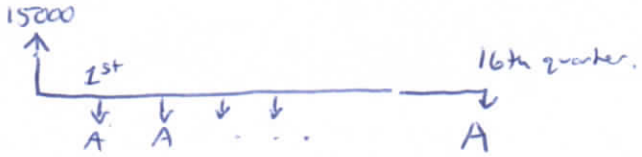
$$1 + \frac{r}{12} = 5^{\frac{1}{60}}$$

$$\frac{r}{12} = 5^{\frac{1}{60}} - 1$$

$$r = 12\left(5^{\frac{1}{60}} - 1\right) = .3262$$

16) $K=4$

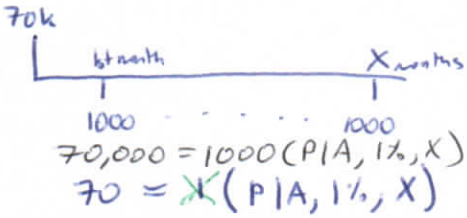
(d) $i_q = e^{\frac{.095}{4}} - 1 \approx .02403$



$15000 = A\underbrace{(P|A, .02403, 16)}_{\text{formula: } 13.541}$

$A = 1140.33$

17) (b)



$i_m = \frac{.12}{12} = .01$

$70,000 = 1000(P|A, 1\%, X)$

$70 = X(P|A, 1\%, X)$

$70 = \frac{(1+i)^N - 1}{i(1+i)^N}$

$= \frac{1}{i} - \frac{1}{i}(1+i)^{-N}, \; i = .01$

$70 = 100(1-(1.01)^{-N})$

$.7 = 1-(1.01)^{-N}$

$(1.01)^{-N} = .3$

$-n\ln 1.01 = \ln .3$

$n = -\frac{\ln .3}{\ln 1.01}$

$n \approx 121$

$\Rightarrow \boxed{X \approx 121}$

18) (a)

monthly payments at 3.5% APR (monthly) $= 22,600\underbrace{(A|P, \frac{.035}{12}, 36)}_{.02930}$

$i_m = \frac{.035}{12} = .0029167$

$= 662.23$ /monthly for 36 months.

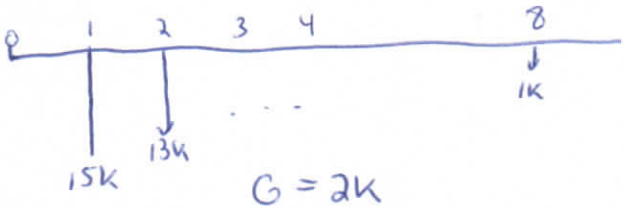
$PW = 662.23(P|A, \overset{.005}{\frac{.06}{12}}, 36) = \$21768.16$ (OPTION 2)

$21768.16 - 20,500 = \$1268.16$

(OPTION 2) − (OPTION ONE)

19) (b)

$i_a = \left(1 + \frac{.09}{12}\right)^{12} - 1 = .09381$



$G = 2K$

$15K - 2K\overbrace{(A|G, .09381, 8)}^{3.0333} = A$

$A = \$8933.40$

20) (e)

$i_s = \left(1 + \frac{.08}{4}\right)^2 - 1 = .0404$

$i_a = \left(1 + \frac{.08}{4}\right)^4 - 1 = .08243$

$V_{15} = A\overbrace{(F|A, .0404, 30)}^{56.4612} = 30k\overbrace{(P|A, .08243, 15)}^{8.4339}$

$A = \$4481.27$

$\frac{(1.0404)^{30} - 1}{.0404}$

$\frac{(1.08243^{15} - 1)}{.08243(1.08243)^{15}}$

(e)